THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET A

Chapter 4: **Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Three angles of a quadrilateral are 80°, 100° and 90°. The fourth angle is: 1

(a) 80° **(b)** 110° **(c)** 100° **(d)** 90°

2. Three angles of a quadrilateral are 50°, 130° and 90°. The fourth angle is: 1

(a) 110° **(b)** 80° **(c)** 100° **(d)** 90°

3. One angle of a parallelogram is 108°. Its adjacent angle is: 1

(a) 18° **(b)** 72° **(c)** 108° **(d)** 144°

4. The diagonals of a rhombus are 16 cm and 30 cm. Its side is: 1

(a) 16 cm **(b)** 17 cm **(c)** 18 cm **(d)** 23 cm

5. The diagonal of a rectangle with sides 9 cm and 12 cm is: 1

(a) 14 cm **(b)** 16 cm **(c)** 21 cm **(d)** 15 cm

6. Diagonals that bisect each other at right angles but are unequal belong to a: 1

(a) trapezium **(b)** rhombus **(c)** kite **(d)** rectangle

7. In a kite, the diagonals: 1

(a) bisect each other **(b)** are equal

(c) meet at right angles **(d)** are parallel

8. The diagonals of a parallelogram: 1

(a) are equal **(b)** are parallel

(c) are perpendicular **(d)** bisect each other

9. An isosceles trapezium has: 1

(a) equal non-parallel sides **(b)** perpendicular diagonals

(c) four right angles **(d)** equal parallel sides

10. A diagonal of a rectangle divides it into: 1

(a) two parallelograms **(b)** two equilateral triangles

(c) three triangles **(d)** two congruent right triangles

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. State the angle sum property of a quadrilateral. 2

12. Three angles of a quadrilateral are 75°, 105° and 90°. Find the fourth. 2

13. List two properties of the diagonals of a rectangle. 2

14. List two properties of the diagonals of a rhombus. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. The angles of a quadrilateral are in the ratio 2:3:4:3. Find all the angles. 3

16. The angles of a quadrilateral are in the ratio 1:2:3:4. Find all the angles. 3

17. The angles of a quadrilateral are in the ratio 3:4:5:6. Find all the angles. 3

OR

The angles of a quadrilateral are in the ratio 2:2:3:3. Find all the angles. 3

18. The angles of a quadrilateral are in the ratio 4:5:6:5. Find all the angles. 3

OR

The angles of a quadrilateral are in the ratio 1:3:4:7. Find all the angles. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Prove that the sum of the angles of a quadrilateral is 360°. 5
(ii) Find the fourth angle if three are 85°, 95° and 70°.

OR

(i) List the properties of a parallelogram. 5
(ii) In parallelogram PQRS, $\angle P = (2x + 20)^\circ$, $\angle Q = (3x - 10)^\circ$. Find x and all angles.

20. (i) Compare a rhombus and a rectangle on sides, angles and diagonals. 5
(ii) State the quadrilateral that is both.

OR

Case study: A kite-maker uses two sticks of length 60 cm and 40 cm placed perpendicular so that the shorter stick is bisected. 5
(i) Name the shape.
(ii) Find the area of paper.
(iii) Is it a rhombus?

END OF QUESTION PAPER — Total: 40 Marks

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